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DriveIQ-Intelligent Report Generation System for BMW Sales Analytics

Project Domain

Power BI with AI Chatbot

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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Year - 2026

3. Problem Statement

The global automotive industry, represented by companies such as BMW, generates massive volumes of sales and vehicle-related data across multiple regions, models, and customer segments. This data includes information such as vehicle specifications, pricing, sales volume, fuel types, and regional preferences. However, this data is often complex, fragmented, and not readily structured for strategic decision-making.

One of the **major challenges in the automotive domain is transforming raw and unorganized data into meaningful business insights**. Without proper data cleaning, transformation, and visualization, organizations struggle to identify key patterns such as top-performing models, regional demand variations, customer preferences, and emerging trends like electric vehicle adoption. As a result, valuable insights remain hidden, leading to missed opportunities in production planning, market targeting, and revenue optimization.

Furthermore, with the rise of Artificial Intelligence tools like Microsoft Copilot and AI-powered chatbots, there is a growing opportunity to **automate insight generation and reporting**. However, a significant gap exists in effectively utilizing these tools due to a lack of prompt engineering knowledge. Many users are unable to extract accurate, contextual, and business-relevant outputs from AI systems, limiting their practical impact in real-world decision-making.

This problem is highly significant in today's data-driven world, where organizations must make fast and informed strategic decisions to remain competitive. In the automotive sector, incorrect or delayed insights can directly impact sales performance, inventory management, and customer satisfaction.

This project aims to address these challenges by developing a comprehensive Business Intelligence solution using Power BI. It focuses on transforming raw BMW global sales data (2010–2024) into structured, interactive dashboards that provide deep insights across multiple dimensions such as market performance, customer preferences, technology trends, and strategic production planning.

Additionally, the project integrates AI through chatbot-based report generation using prompt engineering techniques, enabling automated and intelligent summarization of insights. This bridges the gap between data analysis and decision-making by making insights more accessible and actionable.

The primary stakeholders affected by this problem include business analysts, sales managers, strategic planners, and decision-makers in the automotive industry. By providing a structured BI dashboard combined with AI-powered reporting, the

project delivers value in the form of improved decision-making, enhanced operational efficiency, and the ability to quickly adapt to changing market trends.

4. Objectives

The primary objectives of this project are designed to transform raw automotive data into actionable business insights using Business Intelligence and Artificial Intelligence techniques. The specific goals are as follows:

- To build a smart decision-support system that helps identify optimal vehicle configurations (engine size, fuel type, transmission, and color) for different regions, enabling better production planning and market targeting for BMW.
- To design and develop a **multi-page interactive Power BI dashboard** that effectively visualizes:
    - Market performance insights
    - Customer preferences
    - Technology and performance trends
    - Market segmentation
    - Strategic production insights
    - Advanced multivariate analysis
- To identify key business insights such as top-performing BMW models, regional dominance, electric vehicle adoption trends, and revenue vs volume trade-offs to support strategic decision-making.
- To integrate Artificial Intelligence using tools like Microsoft Copilot and chatbot systems for **automated report generation**, enabling users to generate insights and summaries using well-structured prompts.
- To develop a system that bridges the gap between raw data and decision-making by combining Business Intelligence dashboards with AI-powered narrative generation.

5. Methodology / Implementation Plan

This project follows a structured five-phase workflow to transform raw automotive data into actionable insights and AI-powered reports using Business Intelligence techniques.

#	Phase	Description
1	**Data Collection & Exploration**	The dataset used in this project is the BMW Global Sales Dataset (2010–2024) from Kaggle, representing the automotive domain of BMW. The dataset includes attributes such as model, year, region, fuel type, transmission, engine size, price, and sales volume.

		Initial exploration is performed in Power BI to: • Understand data structure, column types, and relationships • Identify total records and distribution across years and regions • Detect inconsistencies, null values, and anomalies This phase establishes a strong foundation for further data processing and analysis.
2	**Data Cleaning & Preprocessing**	Data cleaning and transformation are carried out using Power BI Query Editor to ensure data quality and consistency. Key steps include: • Removing duplicate records • Handling missing/null values • Standardizing formats (text, numeric, categorical fields) • Correcting inconsistent entries in columns like fuel type and transmission Additionally, **custom columns** are created to enhance analysis: • Revenue (Price × Sales Volume) • Vehicle Type classification (SUV, Sedan, etc.) • Claimed Mileage (derived logically based on engine/fuel type) • Development Nation (Developed, Developing, Mixed) This phase ensures the dataset is structured, reliable, and analysis-ready.
3	**Analysis & AI Integration**	Advanced data analysis is performed using Power BI with DAX measures and calculated fields to extract meaningful insights. Key analytical approaches include: • KPI calculations using DAX Measure (Total Revenue, Total Sales Volume, Growth Rate) • Comparative analysis (Revenue vs Volume, Region-wise performance)

		• Trend analysis (Year-over-year growth, EV adoption trends) • Segmentation (Customer preferences, vehicle types, development levels) To enhance the project, AI tools such as Microsoft Copilot and chatbot integration (via Copilot Studio) are used to: • Generate automated summaries and insights • Convert dashboard data into natural language reports • Enable users to interact with the data using prompts Prompt engineering techniques are applied to ensure accurate and context-aware AI outputs.
4	**Visualization / Output**	A **six-page interactive Power BI dashboard** is developed to present insights in a structured and user-friendly manner: • **Page A:** Market Performance Insights • **Page B:** Customer Preference Insights • **Page C:** Technology & Performance Insights • **Page D:** Market Segmentation Insights • **Page E:** Strategic Production Insights • **Page F:** Advanced Multivariate Insights The dashboard includes: • KPI cards (Revenue, Sales Volume, Top Model, Top Region) • Charts (Bar, Line, Scatter, Treemap, Matrix) • Slicers (Year, Region, Fuel Type, Vehicle Type) • Drill-down and cross-filtering features Additionally, an AI chatbot interface is integrated to generate dynamic reports and summaries based on user queries.
5	**Documentation & Submission**	The complete project is documented systematically, covering: • Problem statement, objectives, and methodology • Data cleaning steps and transformation logic • Dashboard design and insights explanation

		• AI integration and prompt usage All project files, including Power BI dashboard (.pbix), dataset, and documentation, are uploaded to GitHub. A demonstration video is recorded to showcase: • Dashboard functionality • Key insights • AI chatbot-based report generation The final submission includes the GitHub repository link, video demonstration, and dashboard output for evaluation.

6. Resources and Timeline

A. Hardware & Software Requirements

The successful implementation of this project requires the following hardware and software resources:

1. **Hardware Requirements:**
   - Laptop or PC with Intel Core i3 or higher processor
   - Minimum 4 GB RAM (8 GB recommended for smooth Power BI performance)
   - Sufficient storage for dataset, Power BI files, and project documentation
2. **Software Requirements:**
   - Operating System: Windows 10 / 11 or macOS
   - Microsoft Excel (Microsoft 365 / Excel 2019 or later) — for initial data understanding and support analysis
   - Power BI Desktop (latest version) — for data cleaning, transformation, and dashboard development
   - Microsoft Copilot — for AI-assisted insights and report generation
   - AI Chatbot tools such as ChatGPT, Google Gemini, or Copilot Web — for prompt engineering and automated content generation
   - Web browser (latest version of Google Chrome or Microsoft Edge)
   - Stable Internet connection — required for AI tools, Copilot Studio integration, and GitHub upload

These tools collectively enable data processing, visualization, and AI-powered report generation for the BMW sales analysis project.

B. Dataset

The dataset used in this project is a **pre-existing, publicly available dataset** focused on global automotive sales of BMW, covering the time period from 2010 to 2024.

- **Source:**
  The dataset is obtained from a public repository, Kaggle.
- **Dataset Description:**
  The dataset includes the following key attributes:
  - Model (Vehicle model name)
  - Year (2010–2024)
  - Region (Geographical sales market)
  - Color (Vehicle color options)
  - Fuel Type (Petrol, Diesel, Hybrid, etc.)
  - Transmission (Manual/Automatic)
  - Engine Size (in liters)
  - Odometer Reading
  - Price (in USD)
  - Sales Volume
  - Sales Classification (High/Low)

  In addition to the original dataset, several **derived columns** are created during preprocessing to enhance analysis:
  - Revenue (Price × Sales Volume)
  - Vehicle Type (SUV, Sedan, Performance Sedan, etc.)
  - Claimed Mileage (KM/L)
  - Development Nation (Developed, Developing, Mixed)
- **Use in AI Integration:**
  The dataset is used as the primary input for generating insights, which are further processed using AI tools such as Microsoft Copilot and chatbot systems. These tools utilize structured prompts to convert dashboard insights into automated reports and summaries, supporting the project's objective of AI-driven data storytelling.

C. Project Timeline

The project is planned to be completed in a structured timeline over **4–6 weeks**:

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*

2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

This project aims to deliver a comprehensive Business Intelligence solution integrated with Artificial Intelligence to transform raw automotive data into meaningful insights and actionable reports. The expected outcomes are as follows:

- A clean and well-structured BMW global sales dataset (2010–2024) with all preprocessing steps clearly documented, including handling of missing values, standardization, and creation of derived columns, along with a clear before-and-after comparison.
- A fully functional **six-page interactive Power BI dashboard** that effectively presents insights across market performance, customer preferences, technology trends, market segmentation, strategic production, and advanced multivariate analysis for BMW.
- Clear identification of key business insights such as:
  1. Top-performing models by revenue and sales volume
  2. Region-wise market dominance and preferences
  3. Growth trends and electric vehicle adoption
  4. Revenue vs volume trade-offs and premium market segments
- A GitHub repository containing:
  1. Power BI dashboard file (.pbix)
  2. Cleaned dataset
  3. Project report (synopsis + final documentation)
  4. AI prompts and generated outputs
- A recorded video demonstration showcasing, Dashboard navigation and key insights and working of AI chatbot integration

8. References

List all references, tutorials, and documentation used in the project. Students must cite any material they referred to, including research papers, GitHub repositories, YouTube tutorials, blogs, and official documentation.

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning
- Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel
- Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/
- Microsoft Corporation. Microsoft Copilot. Available at: https://copilot.microsoft.com
- Kaggle Inc. Kaggle. *BMW Global Sales Dataset (2010–2024)*. Available at: https://www.kaggle.com/datasets/ayeshaseherr/bmw-dataset